Polarization of Gravitational Waves From Turbulence: Tension Between Simulation and Analytic Approaches

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I. KRAICHNAN–K41 MODEL (GOGOBERIDZE 2007)

We adopt the spatial Fourier pair

$$R_{ij}(\mathbf{r},\tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k},\tau), \qquad F_{ij}(\mathbf{k},\tau) = \int d^3r\, e^{-i\mathbf{k}\cdot\mathbf{r}} R_{ij}(\mathbf{r},\tau), \tag{1}$$

and the temporal Fourier pair

$$f(\eta_k,\tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k,\omega), \qquad \tilde{f}(\eta_k,\omega) = \int d\tau\, e^{i\omega\tau} f(\eta_k,\tau). \tag{2}$$

Starting with all the definitions from Eqs. 23–26 of [1],

$$\begin{aligned} R_{ij}(\mathbf{r},\tau) &= \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k},\tau) \\ &= \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} A(k) P_{ij}(\hat{k}) f(\eta_k,\tau) \\ &= \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} A(k) P_{ij}(\hat{k}) \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k,\omega). \end{aligned} \tag{3}$$

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Here $A(k) = C_k\epsilon^{2/3}k^{-11/3}/4\pi$, $P_{ij}(\hat{k}) = \left(\delta_{ij} - \frac{k_i k_j}{k^2}\right)$, $f(\eta_k,\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}}\epsilon^{1/3}k^{2/3}$. Equation 28 of [1] gives

$$R_{ijij}(\mathbf{x}', \mathbf{x}'+\mathbf{r}, \tau) = R_{ii}(\mathbf{r},\tau)R_{jj}(\mathbf{r},\tau) + \frac{1}{3}R_{ij}(\mathbf{r},\tau)R_{ij}(\mathbf{r},\tau). \tag{4}$$

$$H_{ijkl}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^4}\int d\tau d^3\boldsymbol{\xi}\, e^{i\omega\tau}e^{-i\mathbf{k}\cdot\boldsymbol{\xi}}R_{ijkl}(\boldsymbol{\xi},\tau). \tag{5}$$

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^4}\int d\tau d^3\mathbf{r}\, e^{i\omega\tau}e^{-i\mathbf{k}\cdot\mathbf{r}}\left(R_{ii}(\mathbf{r},\tau)R_{jj}(\mathbf{r},\tau) + \frac{1}{3}R_{ij}(\mathbf{r},\tau)R_{ij}(\mathbf{r},\tau)\right). \tag{6}$$

$$\begin{aligned} H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^4}\int d\tau d^3\mathbf{r}\, e^{i\omega\tau}e^{-i\mathbf{k}\cdot\mathbf{r}}\int \frac{d^3q}{(2\pi)^3}\frac{d^3p}{(2\pi)^3}e^{i(\mathbf{q}+\mathbf{p})\cdot\mathbf{r}} \\ \times\left[F_{ii}(\mathbf{q},\tau)F_{jj}(\mathbf{p},\tau) + \frac{1}{3}F_{ij}(\mathbf{q},\tau)F_{ij}(\mathbf{p},\tau)\right]. \end{aligned} \tag{7}$$

The $\mathbf{r}$ integral gives

$$\int d^3r\, e^{-i\mathbf{k}\cdot\mathbf{r}}e^{i(\mathbf{q}+\mathbf{p})\cdot\mathbf{r}} = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{q}-\mathbf{p}), \tag{8}$$

so integrating over $\mathbf{p}$ and renaming $\mathbf{q} \to \mathbf{k}_1$, $\mathbf{u} = \mathbf{k} - \mathbf{k}_1$ with $u = |\mathbf{u}|$, we obtain

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7}\int d\tau\, e^{i\omega\tau}\int d^3k_1\left[F_{ii}(\mathbf{k}_1,\tau)F_{jj}(\mathbf{u},\tau) + \frac{1}{3}F_{ij}(\mathbf{k}_1,\tau)F_{ij}(\mathbf{u},\tau)\right]. \tag{9}$$

Next insert the temporal Fourier representation

$$F_{ab}(\mathbf{k},\tau) = \int \frac{d\omega}{2\pi}e^{-i\omega\tau}\tilde{F}_{ab}(\mathbf{k},\omega), \tag{10}$$

which gives

$$\begin{aligned} H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7}\int d^3k_1\int d\tau\, e^{i\omega\tau}\int\frac{d\omega_1}{2\pi}\frac{d\omega_2}{2\pi}e^{-i(\omega_1+\omega_2)\tau} \\ \times\left[\tilde{F}_{ii}(\mathbf{k}_1,\omega_1)\tilde{F}_{jj}(\mathbf{u},\omega_2) + \frac{1}{3}\tilde{F}_{ij}(\mathbf{k}_1,\omega_1)\tilde{F}_{ij}(\mathbf{u},\omega_2)\right]. \end{aligned} \tag{11}$$

The τ integral is

$$\int d\tau\, e^{i\omega\tau}e^{-i(\omega_1+\omega_2)\tau} = 2\pi\delta(\omega-\omega_1-\omega_2), \tag{12}$$

so integrating over ω_2 with $\omega_2 = \omega - \omega_1$ yields the convolution form

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8}\int d^3k_1\int d\omega_1\left[\tilde{F}_{ii}(\mathbf{k}_1,\omega_1)\tilde{F}_{jj}(\mathbf{u},\omega-\omega_1) + \frac{1}{3}\tilde{F}_{ij}(\mathbf{k}_1,\omega_1)\tilde{F}_{ij}(\mathbf{u},\omega-\omega_1)\right]. \tag{13}$$

Now substitute the isotropic incompressible model

$$F_{ij}(\mathbf{k},\tau) = A(k)P_{ij}(\hat{k})f(\eta_k,\tau), \qquad \tilde{F}_{ij}(\mathbf{k},\omega) = A(k)P_{ij}(\hat{k})\tilde{f}(\eta_k,\omega), \tag{14}$$

and evaluate the index contractions explicitly. First,

$$P_{ii}(\hat{k}) = \delta_{ii} - \hat{k}_i\hat{k}_i = 3 - 1 = 2. \tag{15}$$

Second,

$$\begin{aligned}P_{ij}(\hat{k}_1)P_{ij}(\hat{u}) &= (\delta_{ij} - \hat{k}_{1i}\hat{k}_{1j})(\delta_{ij} - \hat{u}_i\hat{u}_j)\\ &= \delta_{ij}\delta_{ij} - \delta_{ij}\hat{u}_i\hat{u}_j - \hat{k}_{1i}\hat{k}_{1j}\delta_{ij} + \hat{k}_{1i}\hat{k}_{1j}\hat{u}_i\hat{u}_j\\ &= 3 - 1 - 1 + (\hat{k}_1\cdot\hat{u})^2 = 1 + (\hat{k}_1\cdot\hat{u})^2.\end{aligned} \tag{16}$$

Using Eqs. (15)–(16) in Eq. (13) gives

$$\begin{aligned}H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8}\int d^3k_1 \int d\omega_1 A(k_1)A(u)\tilde{f}(\eta_{k_1}, \omega_1)\tilde{f}(\eta_u, \omega - \omega_1)\\ \times\left[4 + \frac{1}{3}\left(1 + (\hat{k}_1\cdot\hat{u})^2\right)\right].\end{aligned} \tag{17}$$

This is the Fourier transform valid for all k; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \leq \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

Using $\mathbf{k} = \mathbf{k}_1 + \mathbf{u}$ so that

$$k^2 = k_1^2 + u^2 + 2k_1u(\hat{k}_1\cdot\hat{u}) \quad\Rightarrow\quad \hat{k}_1\cdot\hat{u} = \frac{k^2 - k_1^2 - u^2}{2k_1u}, \tag{18}$$

Eq. (17) becomes

$$\begin{aligned}H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8}\int d^3k_1 \int d\omega_1 A(k_1)A(u)\tilde{f}(\eta_{k_1}, \omega_1)\tilde{f}(\eta_u, \omega - \omega_1)\\ \times\left[4 + \frac{1}{3}\left(1 + \left(\frac{k^2 - k_1^2 - u^2}{2k_1u}\right)^2\right)\right].\end{aligned} \tag{19}$$

Expanding the bracket,

$$\begin{aligned}\left[4 + \frac{1}{3}\left(1 + \left(\frac{k^2 - k_1^2 - u^2}{2k_1u}\right)^2\right)\right] &= \left[4 + \frac{1}{3} + \frac{1}{12k_1^2u^2}\left(k^4 + k_1^4 + u^4 - 2k_1^2k^2 - 2k^2u^2 + 2k_1^2u^2\right)\right]\\ &= \left[\frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2}\right].\end{aligned} \tag{20}$$

Combining, we arrive at (cf. Eq. (29) of [1])

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1\; A(k_1)A(u)\,\tilde{f}(\eta_{k_1}, \omega_1)\,\tilde{f}(\eta_u, \omega - \omega_1) \times \mathcal{K}(k, k_1, u), \tag{21}$$

where we define the geometric kernel

$$\mathcal{K}(k, k_1, u) \equiv \left[\frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2}\right]. \tag{22}$$

The temporal Fourier transform of $f(\eta_k, \tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$ is

$$\tilde{f}(\eta_k, \omega) = \frac{2}{\eta_k}\exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right). \tag{23}$$

Using $A(k) = C_k\epsilon^{2/3}k^{-11/3}/4\pi$ and $\eta_k = \frac{1}{\sqrt{2\pi}}\epsilon^{1/3}k^{2/3}$ from [1], the combined factor is

$$g(k, \omega) \equiv A(k)\,\tilde{f}(\eta_k, \omega) = \frac{2E_k}{4\pi k^2\eta_k}\exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \tag{24}$$

where $E_k = C_k \epsilon^{2/3} k^{-5/3}$. Explicitly introducing integration bounds:

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int_{\mathbb{R}^4} d^3k_1\, d\omega_1\, g(k_1, \omega_1)\, g(u, \omega - \omega_1)\, \mathcal{K}(k, k_1, u). \tag{25}$$

Moving the k_1 integral to spherical coordinates, with $\mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$ the cosine of the angle between $\mathbf{k}_1$ and $\mathbf{k}$:

$$d^3k_1 = k_1^2\, dk_1\, d\Omega_1 = k_1^2\, dk_1\, d\phi\, d\mu, \qquad \mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1 \in [-1, 1],\ \ \phi \in [0, 2\pi), \tag{26}$$

$$u \equiv |\mathbf{k} - \mathbf{k}_1| = \sqrt{k^2 + k_1^2 - 2kk_1\mu}. \tag{27}$$

After integrating over the azimuthal angle ($\int_0^{2\pi} d\phi = 2\pi$):

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty dk_1\, k_1^2 \int_{-1}^1 d\mu \int_{-\infty}^\infty d\omega_1\; g(k_1, \omega_1)\, g(u, \omega - \omega_1)\; \mathcal{K}(k, k_1, u). \tag{28}$$

Changing the angular variable from μ to u using Eq. (27),

$$d\mu = \frac{u\, du}{k\, k_1}, \tag{29}$$

with bounds $u \in [|k - k_1|, k + k_1]$:

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty dk_1\, \frac{k_1}{k} \int_{|k-k_1|}^{k+k_1} du\; u \int_{-\infty}^\infty d\omega_1\; g(k_1, \omega_1)\, g(u, \omega - \omega_1)\; \mathcal{K}(k, k_1, u). \tag{30}$$

Splitting $\mathcal{K}$ into parts with different powers of u to separate the u-integrals:

$$\begin{aligned} H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = {} & \frac{1}{(2\pi)^7} \int_0^\infty dk_1\, \frac{k_1}{k} \int_{-\infty}^\infty d\omega_1\; g(k_1, \omega_1) \times \left(\frac{27}{6} - \frac{k^2}{6k_1^2} \right) \int_{|k-k_1|}^{k+k_1} du\; u\; g(u, \omega - \omega_1) \\ & + \frac{1}{(2\pi)^7} \int_0^\infty dk_1\, \frac{k_1}{k} \int_{-\infty}^\infty d\omega_1\; g(k_1, \omega_1) \times \left(\frac{k^4}{12k_1^2} + \frac{k_1^2}{12} - \frac{k^2}{6} \right) \int_{|k-k_1|}^{k+k_1} du\; \frac{1}{u}\; g(u, \omega - \omega_1) \\ & + \frac{1}{(2\pi)^7} \int_0^\infty dk_1\, \frac{k_1}{k} \int_{-\infty}^\infty d\omega_1\; g(k_1, \omega_1) \times \frac{1}{12k_1^2} \int_{|k-k_1|}^{k+k_1} du\; u^3\; g(u, \omega - \omega_1). \end{aligned} \tag{31}$$

This has the same structure as Eq. (31) of [1].

Introducing the dimensionless variables

$$q \equiv \frac{k_1}{k}, \qquad s \equiv \frac{u}{k}, \qquad \Omega \equiv \frac{\omega}{\epsilon^{1/3} k^{2/3}}, \qquad \Omega_1 \equiv \frac{\omega_1}{\epsilon^{1/3} k^{2/3}}, \tag{32}$$

the integration measures transform as

$$dk_1 = k\, dq, \qquad du = k\, ds, \qquad d\omega_1 = \epsilon^{1/3} k^{2/3}\, d\Omega_1, \tag{33}$$

and the u-integration bounds become

$$u \in \big[|k - k_1|,\ k + k_1\big] \quad \longrightarrow \quad s \in \big[|1 - q|,\ 1 + q\big]. \tag{34}$$

Using the explicit form of $g(k, \omega)$,

$$g(k, \omega) = \frac{1}{\sqrt{2\pi}}\, C_k\, \epsilon^{1/3}\, k^{-13/3} \exp\!\big(-2\Omega^2\big)\,, \tag{35}$$

the product $g(k_1, \omega_1) g(u, \omega - \omega_1)$ factorizes into an overall dimensional prefactor and a purely dimensionless Gaussian,

$$\begin{aligned} g(k_1, \omega_1) g(u, \omega - \omega_1) = {} & \frac{1}{2\pi}\, C_k^2\, \epsilon^{2/3}\, k^{-26/3}\, q^{-13/3} s^{-13/3} \\ & \times \exp\!\Big(-2\Omega_1^2 q^{-4/3}\Big) \exp\!\Big(-2(\Omega - \Omega_1)^2 s^{-4/3}\Big)\,. \end{aligned} \tag{36}$$

Collecting all dimensional factors, the correlator takes the scaling form

$$H_{ijij}(\mathbf{x}', \mathbf{k}, \omega) = \frac{C_k^2\,\epsilon}{256\pi^8}\, k^{-5}\, \mathfrak{H}(\Omega), \tag{37}$$

where the entire dependence on physical scales has been absorbed into the dimensionless frequency Ω, and $\mathfrak{H}$ is given by a fully dimensionless integral.

For the split form of the kernel, this dimensionless function reads

$$\begin{aligned}\mathfrak{H}(\Omega) = &\int_0^\infty dq\, q \int_{-\infty}^{\infty} d\Omega_1\, q^{-13/3} \exp\Big(-2\Omega_1^2 q^{-4/3}\Big)\\ &\times \Bigg[\left(\frac{27}{6} - \frac{1}{6q^2}\right) I_{+1}(q,\Omega,\Omega_1) + \left(\frac{1}{12q^2} + \frac{q^2}{12} - \frac{1}{6}\right) I_{-1}(q,\Omega,\Omega_1) + \frac{1}{12q^2}\, I_{+3}(q,\Omega,\Omega_1)\Bigg],\end{aligned} \tag{38}$$

with the dimensionless s-integrals

$$I_{+1}(q,\Omega,\Omega_1) \equiv \int_{|1-q|}^{1+q} ds\; s^{-10/3} \exp\Big(-2(\Omega-\Omega_1)^2 s^{-4/3}\Big)\,, \tag{39}$$

$$I_{-1}(q,\Omega,\Omega_1) \equiv \int_{|1-q|}^{1+q} ds\; s^{-16/3} \exp\Big(-2(\Omega-\Omega_1)^2 s^{-4/3}\Big)\,, \tag{40}$$

$$I_{+3}(q,\Omega,\Omega_1) \equiv \int_{|1-q|}^{1+q} ds\; s^{-4/3} \exp\Big(-2(\Omega-\Omega_1)^2 s^{-4/3}\Big)\,. \tag{41}$$

Equivalently, without splitting the kernel,

$$\mathfrak{H}(\Omega) = \int_0^\infty dq\, q \int_{-\infty}^{\infty} d\Omega_1 \int_{|1-q|}^{1+q} ds\; s\; \Phi(q,\Omega_1; s, \Omega-\Omega_1)\, \widetilde{\mathcal{K}}(q,s), \tag{42}$$

where $\widetilde{\mathcal{K}}(q,s)$ is the kernel $\mathcal{K}(k,k_1,u)$ expressed in terms of q and s, and

$$\Phi(q,\Omega_1; s, \Omega-\Omega_1) \equiv q^{-13/3} s^{-13/3} \exp\Big(-2\Omega_1^2 q^{-4/3}\Big) \exp\Big(-2(\Omega-\Omega_1)^2 s^{-4/3}\Big)\,. \tag{43}$$

In this form, all integrals are manifestly dimensionless, and the dependence on the turbulent energy injection rate ϵ and the wavenumber k is fully contained in the overall prefactor $C_k^2 \epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Eq. A.1 of [1]:

$$\int_0^\infty dy\;\; \exp\left(-Ay^2\right)\;\exp\left(-A(y-x)^2\right) = \left(\frac{\pi}{4(A+B)}\right)^{1/2} \exp\left(\frac{-ABx^2}{A+B}\right) \operatorname{erfc}\left(\frac{-Bx}{\sqrt{A+B}}\right). \tag{44}$$

$$\begin{aligned}H_{ijij}(k,\omega) = &\frac{\varepsilon}{24\pi(2\pi)^{3/2}k} \int_{k_0}^{k_d} dk_1 k_1^{-10/3} \int_{\max(|k_1-k|,k_0)}^{\min(k_1+k,k_d)} du u^{-10/3} \left(k_1^{-4/3} + u^{-4/3}\right)^{-1/2}\\ &\times \left[27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2}\right]\\ &\times \exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) \operatorname{erfc}\left(-\frac{2^{1/2}\varepsilon^{-1/3}\omega}{\left(k_1^{-4/3} + u^{-4/3}\right)^{1/2}}\right).\end{aligned} \tag{45}$$

Here k_0 is the largest scale of the turbulence and k_d is the energy damping scale; the nonlinear integration bounds follow from the finite support of E_k.

Claim: This integral is not divergent as $k \to 0$.

We define the dimensionless parameters

$$\frac{\varepsilon}{k_0} = M^3, \qquad \frac{k_d}{k_0} = R^{3/4}, \qquad p = \frac{k}{k_0}, \qquad q = \frac{\omega}{k_0}. \tag{46}$$

$$x = \left(\frac{k_1}{k_0}\right)^{-4/3}, \qquad y = \left(\frac{u}{k_0}\right)^{-4/3}, \tag{47}$$

$$k_1 = k_0 x^{-3/4}, \qquad u = k_0 y^{-3/4}. \tag{48}$$

$$dk_1 = -\frac{3}{4} k_0 x^{-7/4} dx, \qquad du = -\frac{3}{4} k_0 y^{-7/4} dy. \tag{49}$$

$$k_1^{-10/3} = k_0^{-10/3} x^{5/2}, \qquad u^{-10/3} = k_0^{-10/3} y^{5/2}, \tag{50}$$

$$dk_1\, k_1^{-10/3} = -\frac{3}{4} k_0^{-7/3} x^{3/4} dx, \qquad du\, u^{-10/3} = -\frac{3}{4} k_0^{-7/3} y^{3/4} dy. \tag{51}$$

$$k_1^{-4/3} + u^{-4/3} = k_0^{-4/3}(x+y), \qquad \left(k_1^{-4/3} + u^{-4/3}\right)^{-1/2} = k_0^{2/3}(x+y)^{-1/2}. \tag{52}$$

$$k_1 \in [k_0, k_d] \quad \Longrightarrow \quad x \in [R^{-1}, 1]. \tag{53}$$

$$\tilde{k}_1 = \frac{k_1}{k_0} = x^{-3/4}, \qquad \tilde{u} = \frac{u}{k_0} = y^{-3/4}, \tag{54}$$

$$\tilde{u}_{\min} = \max(|\tilde{k}_1 - p|, 1), \qquad \tilde{u}_{\max} = \min(\tilde{k}_1 + p, R^{3/4}), \tag{55}$$

$$y_{\min} = \tilde{u}_{\max}^{-4/3}, \qquad y_{\max} = \tilde{u}_{\min}^{-4/3}. \tag{56}$$

$$27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2 u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2} = 27 - p^2 x^{3/2} - p^2 y^{3/2} + \frac{p^4}{2} x^{3/2} y^{3/2} + \frac{1}{2} x^{-3/2} y^{3/2} + \frac{1}{2} y^{-3/2} x^{3/2}. \tag{57}$$

$$\exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) = \exp\left(-\frac{2xy}{x+y}\frac{q^2}{M^2}\right), \tag{58}$$

$$\operatorname{erfc}\left(-\frac{2^{1/2}\varepsilon^{-1/3}\omega}{(k_1^{-4/3} + u^{-4/3})^{1/2}}\right) = \operatorname{erfc}\left(-\frac{\sqrt{2}\,q}{M\sqrt{x+y}}\right). \tag{59}$$

$$\begin{aligned} H_{ijij}(p,q) = {} & \frac{3M^3 k_0^{-4}}{256(2\pi)^{3/2} p} \int_{R^{-1}}^{1} dx\, x^{3/4} \int_{y_{\min}(x)}^{y_{\max}(x)} dy\, y^{3/4} (x+y)^{-1/2} \\ & \times \left[27 - p^2 x^{3/2} - p^2 y^{3/2} + \frac{p^4}{2} x^{3/2} y^{3/2} + \frac{1}{2} x^{-3/2} y^{3/2} + \frac{1}{2} y^{-3/2} x^{3/2}\right] \\ & \times \exp\left(-\frac{2xy}{x+y}\frac{q^2}{M^2}\right) \operatorname{erfc}\left(-\frac{\sqrt{2}\,q}{M\sqrt{x+y}}\right). \end{aligned} \tag{60}$$

$$y_{\max} = \min(|x^{-3/4} - p|^{-4/3}, 1), \qquad y_{\min} = \max((x^{-3/4} + p)^{-4/3}, R^{-1}). \tag{61}$$

II. MONOCHROMATIC AND WHITE-NOISE SPECTRA

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function) distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k-space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. 1: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω, (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u = |\mathbf{k} - \mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

A. Monochromatic energy spectrum $E(k) = E_0\delta(k - k_0)$

We set

$$E(k) = E_0\,\delta(k - k_0), \qquad A(k) = \frac{E(k)}{4\pi k^2} = \frac{E_0}{4\pi k^2}\,\delta(k - k_0), \tag{62}$$

so the spectral tensor is

$$F_{ij}(\mathbf{k}, \tau) = \frac{E_0\,\delta(k - k_0)}{4\pi k^2}\,P_{ij}(\hat{k})\,f(\tau), \qquad \tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0\,\delta(k - k_0)}{4\pi k^2}\,P_{ij}(\hat{k})\,\tilde{f}(\omega), \tag{63}$$

where $f(\tau)$ (and its Fourier transform $\tilde{f}(\omega)$) carries the temporal decorrelation, to be specified below. The total kinetic energy density is $\int_0^\infty E(k)\,dk = E_0$.

Starting from Eq. (17) and writing $d^3k_1 = k_1^2\,dk_1\,d\phi\,d\mu$ with $\mu \equiv \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$, the radial k_1 integral immediately yields

$$\int_0^\infty k_1^2\,dk_1\,A(k_1) = \int_0^\infty dk_1\,\frac{E_0\,\delta(k_1 - k_0)}{4\pi} = \frac{E_0}{4\pi}. \tag{64}$$

With k_1 fixed to k_0, the second spectral factor evaluates on $u \equiv |\mathbf{k} - \mathbf{k}_1| = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$:

$$A(u) = \frac{E_0\,\delta(u - k_0)}{4\pi u^2}. \tag{65}$$

We change the angular variable from μ to u using Eq. (27), i.e. $u\,du = kk_0\,|d\mu|$, and perform the azimuthal integral ($\int_0^{2\pi} d\phi = 2\pi$):

$$2\pi \int_{-1}^{1} d\mu\,\frac{E_0\,\delta(u(\mu) - k_0)}{4\pi u(\mu)^2} = \frac{E_0}{2} \int_{|k-k_0|}^{k+k_0} \frac{du}{kk_0}\,\frac{\delta(u - k_0)}{u} = \frac{E_0}{2kk_0^2}\,\Theta(2k_0 - k). \tag{66}$$

The Heaviside factor $\Theta(2k_0 - k)$ enforces the triangle inequality: the root $u = k_0$ lies in the range $[|k - k_0|, k + k_0]$ only when $k \le 2k_0$, i.e. $p \le 2$ with $p \equiv k/k_0$.

The geometric kernel at $k_1 = u = k_0$ evaluates as (using $p = k/k_0$)

$$\mathcal{K}(k, k_0, k_0) = \frac{27}{6} + \frac{k^4}{12k_0^4} + \frac{k_0^2}{12k_0^2} + \frac{k_0^2}{12k_0^2} - \frac{k^2}{6k_0^2} - \frac{k^2}{6k_0^2} = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12} \equiv \mathcal{K}_0(p). \tag{67}$$

Collecting Eqs. (64)–(67) in Eq. (17),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2\,\Theta(2k_0 - k)}{2(2\pi)^8\,kk_0^2}\,\mathcal{K}_0(p)\,\underbrace{\int_{-\infty}^{\infty} d\omega_1\,\tilde{f}(\omega_1)\,\tilde{f}(\omega - \omega_1)}_{I_\omega(\omega)}. \tag{68}$$

All spatial structure has been reduced to the explicit prefactor; the remaining task is to evaluate the temporal convolution I_ω.

1. Kraichnan decorrelation

The Kraichnan model gives $\tilde{f}(\omega) = (2/\eta_0)\exp(-\omega^2/\pi\eta_0^2)$ with the fixed rate $\eta_0 \equiv \eta_{k_0}$. The convolution of two identical Gaussians is itself Gaussian:

$$\begin{aligned} I_\omega(\omega) &= \frac{4}{\eta_0^2}\int_{-\infty}^{\infty} d\omega_1\, \exp\left(-\frac{\omega_1^2 + (\omega-\omega_1)^2}{\pi\eta_0^2}\right) \\ &= \frac{4}{\eta_0^2}\, e^{-\omega^2/(2\pi\eta_0^2)}\int_{-\infty}^{\infty} d\omega_1\, \exp\left(-\frac{2(\omega_1-\omega/2)^2}{\pi\eta_0^2}\right) \\ &= \frac{2\sqrt{2}\pi}{\eta_0}\, \exp\left(-\frac{\omega^2}{2\pi\eta_0^2}\right). \end{aligned} \tag{69}$$

Defining the dimensionless frequency $\Omega \equiv \omega/\eta_0$ and substituting into Eq. (68),

$$H_{ijij}(\mathbf{k},\omega) = \frac{\sqrt{2}\,E_0^2}{(2\pi)^7\, k k_0^2\, \eta_0}\, \mathcal{K}_0(p)\,\Theta(2-p)\, e^{-\Omega^2/(2\pi)}. \tag{70}$$

Writing $k = pk_0$, the entire result separates into a dimensional prefactor times a dimensionless expression:

$$\boxed{\mathfrak{H}^{(\delta k)}_{\text{Kraichnan}}(p,\Omega) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\, e^{-\Omega^2/(2\pi)},} \tag{71}$$

with $\mathcal{K}_0(p) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}$ from Eq. (67). There is no remaining integral: both delta functions have been evaluated in closed form.

2. Decay model

For the decaying-turbulence model the temporal kernel is

$$\tilde{f}(\omega) = \hat{g}(\omega\tau_1), \qquad \hat{g}(q) = e^{iq}(-iq)^{-5/3}\Gamma\left(\tfrac{1}{3}, -iq\right), \tag{72}$$

where τ_1 is the characteristic decay time at scale k_0. Substituting into Eq. (68) and defining $q = \omega\tau_1$, $q_1 = \omega_1\tau_1$, so that $d\omega_1 = dq_1/\tau_1$:

$$I_\omega(\omega) = \int_{-\infty}^{\infty} d\omega_1\, \hat{g}(\omega_1\tau_1)\,\hat{g}((\omega-\omega_1)\tau_1) = \frac{1}{\tau_1}\int_{-\infty}^{\infty} dq_1\, \hat{g}(q_1)\,\hat{g}(q-q_1). \tag{73}$$

Because $\hat{g}$ has no closed-form convolution, this one-dimensional integral must be retained. Inserting Eq. (73) into Eq. (68),

$$H_{ijij}(\mathbf{k},\omega) = \frac{E_0^2\,\Theta(2k_0-k)}{2(2\pi)^8\, k k_0^2\, \tau_1}\, \mathcal{K}_0(p)\int_{-\infty}^{\infty} dq_1\, \hat{g}(q_1)\,\hat{g}(q-q_1). \tag{74}$$

The dimensionless result is

$$\boxed{\mathfrak{H}^{(\delta k)}_{\text{decay}}(p,q) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\int_{-\infty}^{\infty} dq_1\, \hat{g}(q_1)\,\hat{g}(q-q_1).} \tag{75}$$

The spatial integration is again fully resolved by the two delta functions; only a single one-dimensional frequency integral remains.

B. Real-space delta-function correlations

We now consider a model in which the spatial part of the two-point velocity correlation is a delta function in $\mathbf{r}$:

$$R_{ij}(\mathbf{r},\tau) = v_0^2\,\delta^{(3)}(\mathbf{r})\,P_{ij}\,T(\tau), \tag{76}$$

where v_0^2 is the velocity variance and $T(\tau)$ is the (to-be-specified) temporal decorrelation function. Taking the spatial Fourier transform,

$$F_{ij}(\mathbf{k},\tau) = v_0^2\, P_{ij}(\hat{k})\, T(\tau), \qquad \tilde{F}_{ij}(\mathbf{k},\omega) = v_0^2\, P_{ij}(\hat{k})\, \widetilde{T}(\omega), \tag{77}$$

so the spectral amplitude is k-independent: $A(k) = v_0^2$. This corresponds to an energy spectrum $E(k) \propto k^2$ (blue, ultraviolet-dominated), and in practice the integrals must be cut off between the outer scale k_0 and a dissipation scale k_d.

The tensor contractions proceed identically to Sec. 1 and yield the same kernel $\mathcal{K}(k, k_1, u)$. With $A(k_1) = A(u) = v_0^2$ and the temporal factor $\widetilde{T}(\omega)$ independent of k, Eq. (9) becomes

$$H_{ijij}(\mathbf{k},\omega) = \frac{v_0^4}{(2\pi)^7} \underbrace{\int_{-\infty}^{\infty} d\omega_1\, \widetilde{T}(\omega_1)\, \widetilde{T}(\omega-\omega_1)}_{J(\omega)} \int_{k_0}^{k_d} \frac{k_1}{k}\, dk_1 \int_{\max(|k-k_1|,k_0)}^{\min(k+k_1,k_d)} u\, du\, \mathcal{K}(k,k_1,u). \tag{78}$$

The temporal convolution $J(\omega)$ and the double spatial integral are fully decoupled because $\widetilde{T}$ does not depend on k_1 or u. Introducing dimensionless integration variables

$$z \equiv \frac{k_1}{k_0}, \quad y \equiv \frac{u}{k_0}, \quad p \equiv \frac{k}{k_0}, \quad R^{3/4} \equiv \frac{k_d}{k_0}, \tag{79}$$

so $dk_1 = k_0\, dz$, $du = k_0\, dy$, the spatial integral becomes

$$k_0^2 \int_1^{R^{3/4}} \frac{z}{p}\, dz \int_{\max(|p-z|,1)}^{\min(p+z,\, R^{3/4})} y\, dy\, \widetilde{\mathcal{K}}(p,z,y), \tag{80}$$

where $\widetilde{\mathcal{K}}(p,z,y)$ is the geometric kernel of Eq. (19) written in units of k_0:

$$\widetilde{\mathcal{K}}(p,z,y) = \left[\frac{27}{6} + \frac{p^4}{12z^2y^2} + \frac{z^2}{12y^2} + \frac{y^2}{12z^2} - \frac{p^2}{6y^2} - \frac{p^2}{6z^2}\right]. \tag{81}$$

1. *Kraichnan decorrelation*

For a spatially uncorrelated field there is no scale-dependent sweeping rate; we adopt a fixed decorrelation rate η_0 and set $\widetilde{T}(\omega) = (2/\eta_0)\exp(-\omega^2/\pi\eta_0^2)$. The temporal convolution is identical in form to Eq. (69):

$$J_{\text{Kraichnan}}(\omega) = \frac{2\sqrt{2}\pi}{\eta_0}\, \exp\!\left(-\frac{\omega^2}{2\pi\eta_0^2}\right), \tag{82}$$

which is now a k-independent closed form. Substituting Eqs. (82) and (80) into Eq. (78) and defining $\Omega \equiv \omega/\eta_0$:

$$H_{ijij}(\mathbf{k},\omega) = \frac{\sqrt{2}\, v_0^4\, k_0^2}{(2\pi)^6\, \eta_0}\, e^{-\Omega^2/(2\pi)}\, k_0^2 \int_1^{R^{3/4}} \frac{z}{p}\, dz \int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})} y\, dy\, \widetilde{\mathcal{K}}(p,z,y). \tag{83}$$

The dimensionless integral is

$$\boxed{\mathfrak{H}^{(\delta r)}_{\text{Kraichnan}}(p,\Omega) = e^{-\Omega^2/(2\pi)} \int_1^{R^{3/4}} \frac{z}{p}\, dz \int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})} y\, dy\, \widetilde{\mathcal{K}}(p,z,y).} \tag{84}$$

The frequency dependence is an explicit Gaussian factor, while the shape in p is determined by the two-dimensional wavenumber integral.

2. Decay model

For the decaying-turbulence model, $\widetilde{T}(\omega) = \hat{g}(\omega\tau_1)$ with the same $\hat{g}$ as in Sec. 3.1.2. The temporal convolution is again a one-dimensional integral with no closed form:

$$J_{\rm decay}(\omega) = \int_{-\infty}^{\infty} d\omega_1\, \hat{g}(\omega_1\tau_1)\, \hat{g}((\omega-\omega_1)\tau_1) = \frac{1}{\tau_1}\int_{-\infty}^{\infty} dq_1\, \hat{g}(q_1)\, \hat{g}(q-q_1), \qquad q \equiv \omega\tau_1. \tag{85}$$

Substituting into Eq. (78),

$$H_{ijij}(\mathbf{k},\omega) = \frac{v_0^4\, k_0^2}{(2\pi)^7\, \tau_1}\, \mathfrak{H}^{(\delta r)}_{\rm decay}(p,q), \tag{86}$$

where the fully dimensionless expression is now genuinely three-dimensional:

$$\boxed{\mathfrak{H}^{(\delta r)}_{\rm decay}(p,q) = \int_1^{R^{3/4}} \frac{z}{p}\, dz \int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})} y\, dy\, \widetilde{\mathcal{K}}(p,z,y) \int_{-\infty}^{\infty} dq_1\, \hat{g}(q_1)\, \hat{g}(q-q_1).} \tag{87}$$

Because $J_{\rm decay}$ is independent of z and y (since $\widetilde{T}$ carries no k-dependence), the three integrals are separable: the double wavevector integral and the frequency convolution may be computed independently.

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k-k_0)$	Kraichnan	Eq. (71): closed form, no integral
$E_0\delta(k-k_0)$	Decay	Eq. (75): 1D frequency integral
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (84): 2D wavevector integral
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (87): 2D wavevector $\times$ 1D frequency

[1] G. Gogoberidze, T. Kahniashvili, and A. Kosowsky, Phys. Rev. D **76**, 083002 (2007), arXiv:0705.1733 [astro-ph] .